RESEARCH CLI — SYSTEM ARCHITECTURE

TypeScript monorepo · packages/cli + packages/core · github.com/iechor-research/research-cli

packages/cli

- **research.tsx / TUI** — packages/cli/src
- **commands/** — mcp.ts skills.tsx hooks.tsx
- **config/config.ts** — packages/cli/src/config/config.ts
- **acp/** — acpSession.ts acpCommandHandler.ts

packages/core orchestration

- **config.ts** — packages/core/src/config/config.ts
- **research-config-manager.ts** — packages/core/src/config/
- **modelRouterService.ts** — packages/core/src/routing/
- **routingStrategy.ts** — default/classifier/composite
- **memory.ts / storage.ts** — session + compress

route

TOOLS + MCP

- **workspace tools** — fs / shell packages/core/src/tools
- **MCP** — packages/cli/src/commands/mcp.ts
- **sandbox** — config/sandbox-integration
- **packages/sdk** — programmatic embed
- **packages/a2a-server** — agent protocol
- **vscode-ide-companion** — IDE bridge

ACADEMIC MODULES packages/core/src/tools/research/

- **writing** — outline / LaTeX
- **bibliography** — arXiv / PubMed / BibTeX
- **analysis** — stats / citation net
- **investigation** — experiment codegen
- **submission** — journal matcher

PROVIDERS model-provider-config.ts + /model config

- **Gemini / Vertex**
- **OpenAI**
- **Anthropic**
- **Baidu ERNIE**
- **DeepSeek**
- **Qwen / Groq / Kimi**

CLI TUI → core config/router → provider + tools/MCP/research modules → streamed result